

BUILD WITH AI 2026 · TRACK 2: CLEAN AIR & CLIMATE RESILIENCE

VesperAero

Federated Environmental Forensics & Cross-Border Plume Platform



Citizen Sensing



Gemini AI



Gaussian Physics



NASA Ground-Truth

Turning citizens into a hyper-local pollution sensor network

THE GAP

Major BRICS economic corridors consistently miss localized and transboundary emission spikes due to coarse satellite revisit times.

THE FIX

VesperAero turns citizens into hyper-local pollution sensors, augmented with Google Gemini Multimodal AI, Gaussian plume dispersion physics, and NASA satellite ground-truthing.



Citizen Sensors



Gemini Multimodal AI



Gaussian Plume Physics



NASA Ground-Truthing

Three structural gaps in cross-border emission response



Coarse Spatial Resolution

Fixed municipal sensors miss hyper-local flares.



Cross-Border Paralysis

Drifting smog causes jurisdictional disputes without forensic attribution.



Enforcement Lag

Legacy complaints lack telemetry, delaying response by days.

A 4-stage closed-loop pipeline



Gemini Vision

Google Gemini 1.5 Flash Vision ingests raw optical imagery and real-time meteorology to:



Classify Emission Sources

Stubble, kiln, flare



Score Optical Density

1-10 opacity index



Extract Chemical Signatures

PM2.5, SO2, CO



Generate Enforcement Directives

Actionable, ready to dispatch

Atmospheric Gaussian Physics



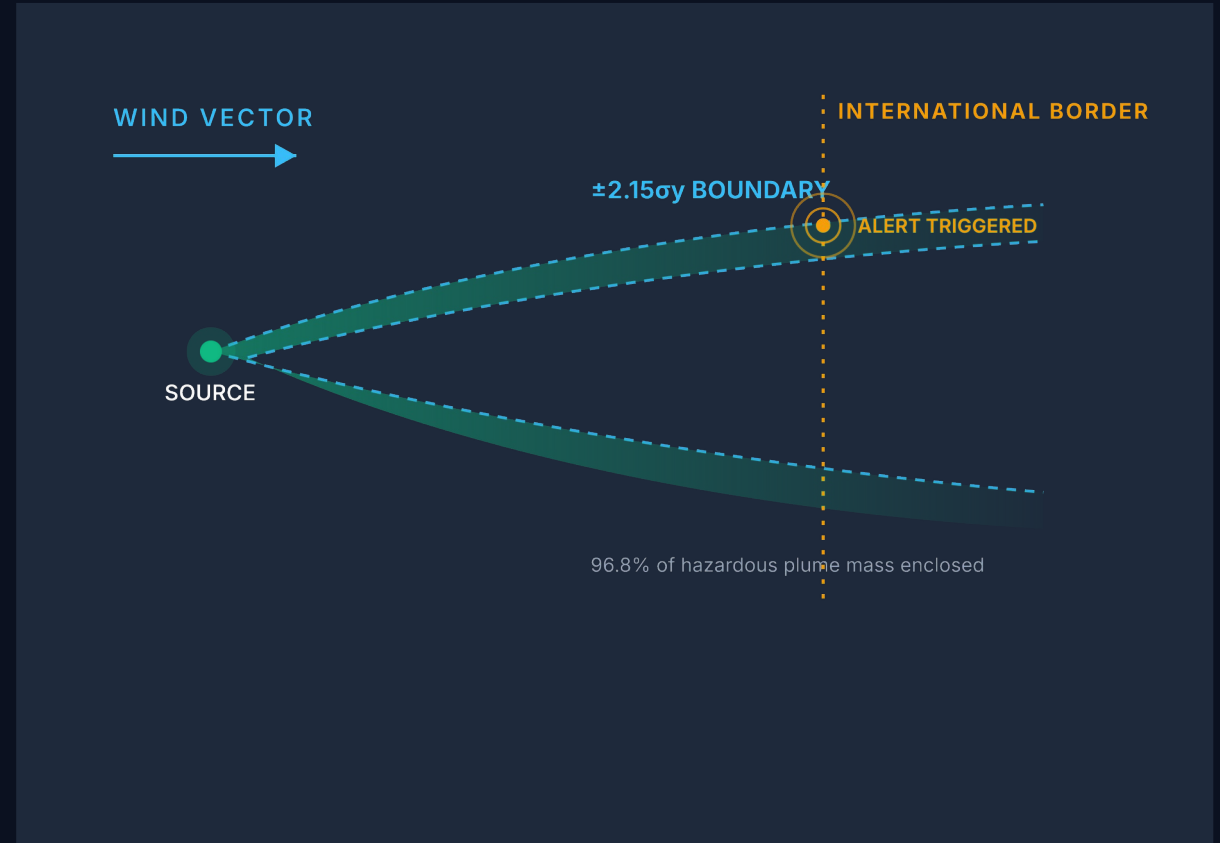
Automatically triggers transboundary alerts when a spatial polygon crosses international borders, enabling proactive rather than reactive public health response.



Integrates Gemini's optical density with Pasquill-Gifford atmospheric dispersion equations, projecting a $\pm 2.15\sigma_y$ boundary that encloses 96.8% of hazardous plume mass downwind based on live wind vectors.



Applies atmospheric stability parameterization across Classes A to F.



NASA FIRMS Orbital Validation



375m

MODIS / VIIRS

Dual-layer verification matches ground reports with orbital infrared signatures.



FRP

RADIATIVE POWER

Dual-layer verification matches ground reports with orbital infrared signatures.



100%

SYNERGY

Dual-layer verification matches ground reports with orbital infrared signatures.

Autonomous Drone Flight Paths

Spatio-temporal clustering filters out complaint noise. For persistent clusters, VesperAero generates autonomous drone waypoints:



Origin

Optical inspection



Plume Core

Gas sampling



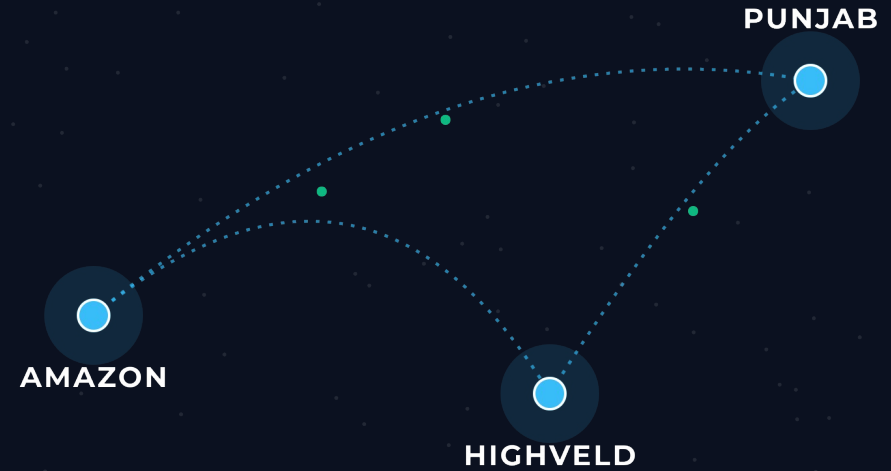
Downwind Boundary

Exposure perimeter

CROSS-BORDER PROTOCOL

BRICS-AERO-FED Protocol

Frictionless data exchange across the Punjab,
Amazon & Highveld corridors.



Built for the field, not the boardroom



Voice Capture



5

5 Languages

English, Hindi, Portuguese,
Russian, and Chinese localization.



100%

Offline APK

Compiled as a production Android
APK with hardware GPS locking.

Metrics



47 / 47

AUTOMATED TESTS

Jest tests passed (100%) across all API routes.



0

SECURITY SCANS

npm security audit & static analysis
— 0 vulnerabilities found.



100%

LIVE ENDPOINTS

Active on Railway Cloud & Netlify
Edge CDN — 100% uptime.

From pilot to protocol



PHASE 1

Municipal Pilots

Deploy in high-density border districts.



PHASE 2

IoT Fusion

Direct telemetry from Continuous Emission Monitors.



PHASE 3

BRICS Grid

Inter-governmental climate treaty data hub.



MISSION

Coordinated Action

Turning transparency into immediate, coordinated climate action.